pfnet / **plamo-embedding-1b**

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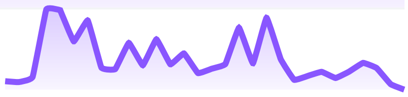

Safetensors

Model size 1.05B params | Tensor type BF16 | Files info

Inference Providers NEW

Sentence Similarity

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Collection including `pfnet/plamo-embedding-1b`

PLaMo Collection
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PLaMo-Embedding-1B

[日本語版のREADME/Japanese README]

Model Overview

PLaMo-Embedding-1B is a Japanese text embedding model developed by Preferred Networks, Inc.

It can convert Japanese text input into numerical vectors and can be used for a wide range of applications, including information retrieval, text classification, and clustering.

As of early April 2025, it achieved top-class scores on JMTEB, a benchmark for Japanese text embedding. It demonstrated particularly outstanding performance, especially in retrieval tasks.

PLaMo-Embedding-1B is released under the Apache v2.0 license, and you can use it freely, including for commercial purposes.

For technical details, please refer to the following Tech Blog post (Ja): https://tech.preferred.jp/ja/blog/plamo-embedding-1b/

Usage

Requirements

```
sentencepiece
torch
transformers
```

Sample Code

```
import torch
import torch.nn.functional as F
from transformers import AutoModel, AutoTokenizer

# You can download models from the Hugging Face Hub 🤗 as follows:
tokenizer = AutoTokenizer.from_pretrained("pfnet/plamo-embedding-1b", trust_remote
model = AutoModel.from_pretrained("pfnet/plamo-embedding-1b", trust_remote_code=Tr

device = "cuda" if torch.cuda.is_available() else "cpu"
model = model.to(device)

query = "PLaMo-Embedding-1Bとは何ですか？"
documents = [
    "PLaMo-Embedding-1Bは、Preferred Networks, Inc. によって開発された日本語テキスト埋め
    "最近は随分と暖かくなりましたね。"
]

with torch.inference_mode():
```

```
    # For embedding query texts in information retrieval, please use the `encode_q
    # You also need to pass the `tokenizer`.
    query_embedding = model.encode_query(query, tokenizer)
    # For other texts/sentences, please use the `encode_document` method.
    # Also, for applications other than information retrieval, please use the `enc
    document_embeddings = model.encode_document(documents, tokenizer)

# The similarity between vectors obtained by inputting sentences into the model is
# This feature can be utilized for applications such as information retrieval.
similarities = F.cosine_similarity(query_embedding, document_embeddings)
print(similarities)
# tensor([0.8812, 0.5533])
```

Note: For `encode_document` and `encode_query`, texts exceeding the model's maximum context length of 4096 will be truncated. Be especially aware that for `encode_query`, a prefix is added internally, making the effective maximum context length slightly shorter.

Benchmarks

We conducted a performance evaluation using JMTEB, a benchmark for Japanese text embedding.

Model	Avg.	Retrieval	STS	Classification	Reranking	Clustering	PairClassificatior
intfloat/multilingual-e5-large	70.90	70.98	79.70	72.89	92.96	51.24	62.15
pkshatech/GLuCoSE-base-ja-v2	72.23	73.36	82.96	74.21	93.01	48.65	62.37
OpenAI/text-embedding-3-large	74.05	74.48	82.52	77.58	93.58	53.32	62.35
cl-nagoya/ruri-large-v2	74.55	76.34	83.17	77.18	93.21	52.14	62.27
Sarashina-Embedding-v1-1B	75.50	77.61	82.71	**78.37**	**93.74**	**53.86**	62.00

Model	Avg.	Retrieval	STS	Classification	Reranking	Clustering	PairClassificatior
PLaMo-Embedding-1B (This model) (*)	**76.10**	**79.94**	**83.14**	77.20	93.57	53.47	62.37

(*): Measured with a context length of 1024. Although the model supports a context length of up to 4096, we measured at 1024 because the context length included during training was up to 1024. However, it is known that evaluating at 4096 does not significantly affect the average score. (Ref: Tech Blog (Ja))。

Model Details

- Model Size: 1B
- Maximum Context Length: 4096 tokens
- Embedding Dimensionality: 2048
- Similarity Function: cosine similarity
- Developer: Preferred Networks, Inc
- Language: Japanese
- License: Apache v2.0

License

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How to cite

```
@online{PLaMoEmbedding1B,
    author    = {Preferred Networks, Inc},
    title     = {PLaMo-Embedding-1B},
    year      = {2025},
    url       = {https://huggingface.co/pfnet/plamo-embedding-1b},
```

```
    urldate    = {2025-04-17}
}
```